Injections versus surjections on cardinals, under excluded middle and the axiom of choice

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:injsurj' (source lines 805-821, Lemma 10.2.9); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The two internal item labels item:cle-inj and item:cle-surj are defined inside the retained statement and used inside the retained proof and statement, so they need no external entry.

Editorial note. The retained counter values (chapter 10, section 2, theorem offset 8) make the counted lemma print as Lemma 10.2.9, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial scope. One original result of the HoTT Book, counted once: the implications between injections and surjections on cardinalities (Lemma 10.2.9).

Allowed standard prerequisites (editorial).

- Injection and surjection. f : A -> B is injective when its fibers are mere propositions, and surjective when every fiber is merely inhabited.. Both notions are the subject of the three implications.
- Excluded middle. For every proposition P, P or not P.. It is assumed for the whole statement and used at source line 825 to decide a fiber.
- Axiom of choice. A surjection merely has a section.. It is assumed in the third part and used at source line 830.

Lemma 10.2.9. *Consider the following statements:*

(i) There is an injection $A \to B$.
(ii) There is a surjection $B \to A$.

Then, assuming excluded middle:

- *Given $a_0 : A$, we have (i)→(ii).*
- *Therefore, if A is merely inhabited, we have (i) → merely (ii).*
- *Assuming the axiom of choice, we have (ii) → merely (i).*

Proof. If $f : A \to B$ is an injection, define $g : B \to A$ at $b : B$ as follows. Since f is injective, the fiber of f at b is a mere proposition. Therefore, by excluded middle, either there is an $a : A$ with $f(a) = b$, or not. In the first case, define $g(b) :\equiv a$; otherwise set $g(b) :\equiv a_0$. Then for any $a : A$, we have $a = g(f(a))$, so g is surjective.

The second statement follows from this by induction on truncation. For the third, if $g : B \to A$ is surjective, then by the axiom of choice, there merely exists a function $f : A \to B$ with $g(f(a)) = a$ for all a. But then f must be injective. □